

Age-Specific Reference Ranges for Total Testosterone in Males: A Systematic Review and Pooled Analysis of 261,985 Men

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Clinicians should use a total testosterone level below 300 ng/dL as a reasonable cut-off in support of the diagnosis of low testosterone.

**Clinicians should adjust
testosterone therapy dosing to
achieve a total testosterone level
in the middle tertile of the normal
reference range.**

But how do we define “Normal?”

CDC Hormone Standardization Program

Framingham Heart Study

European Male Aging Study

Osteoporotic Fractures in Men Study

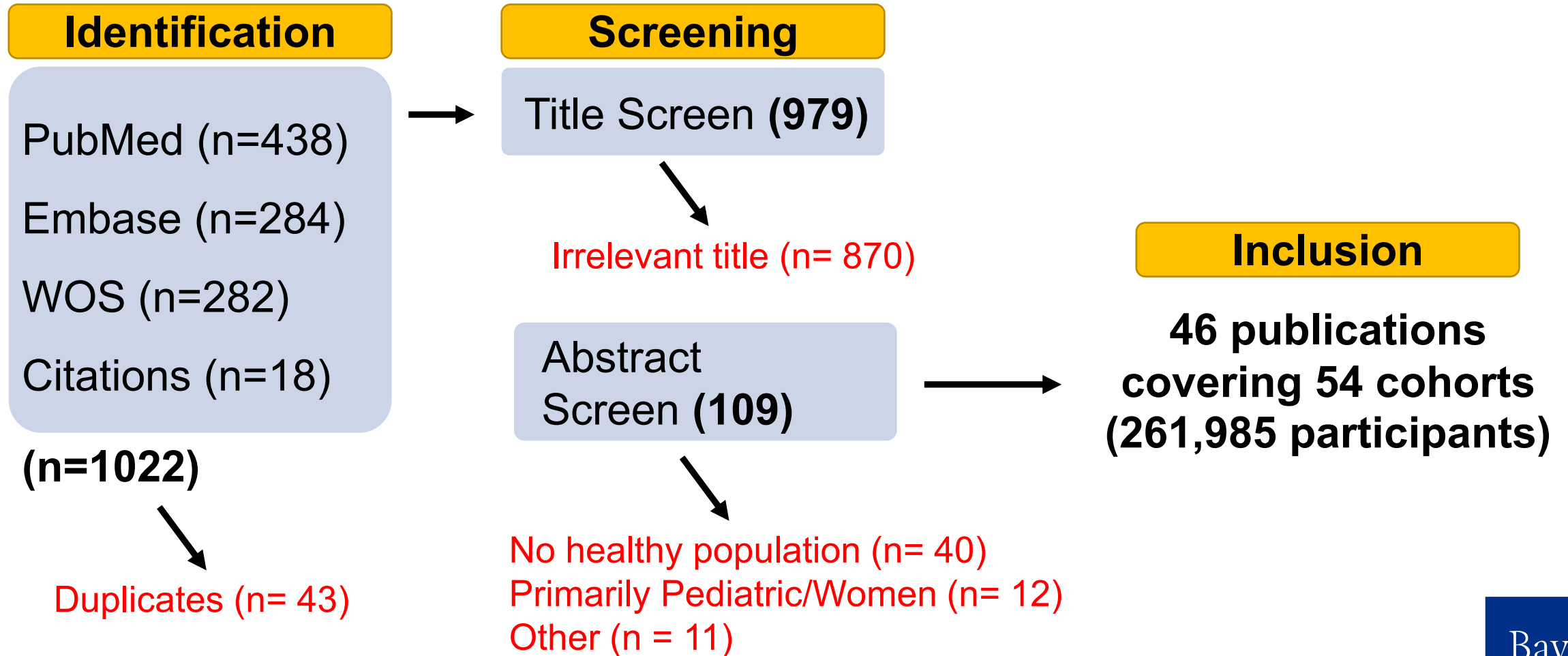
Male Sibling Study of Osteoporosis

**In 9054 with cross-
calibrated LCMS
Testosterone assays**

264 to 916 ng/DL

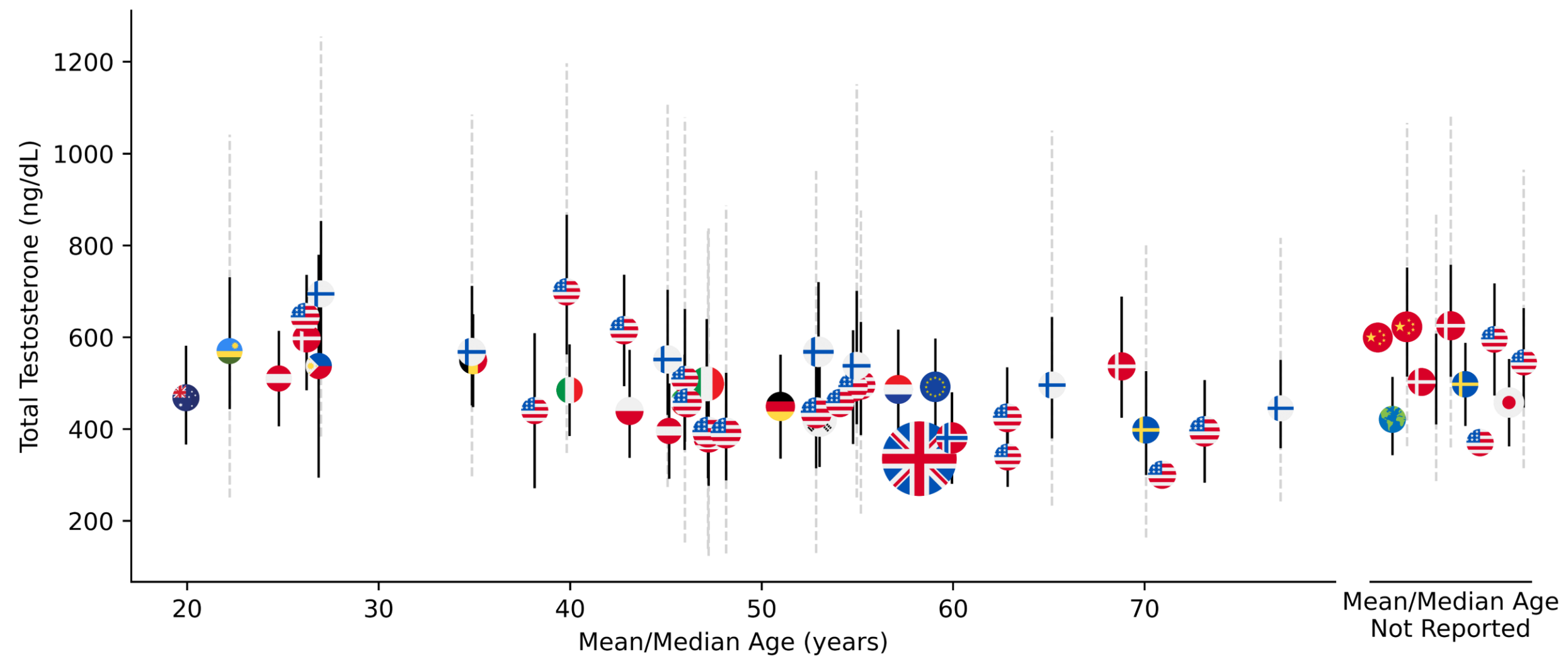
In 19-39 y.o. men

Systematic Review



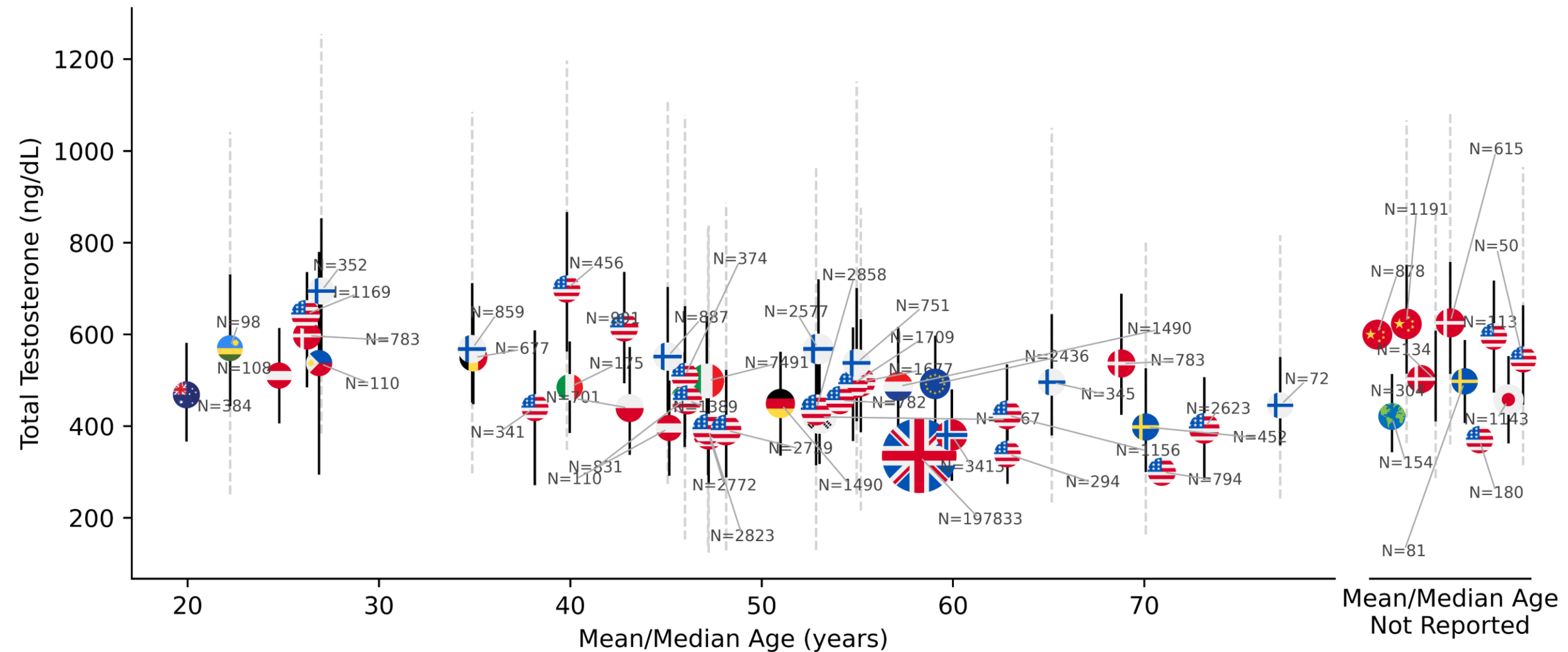
Mean Total T by Age

With T-Ranges



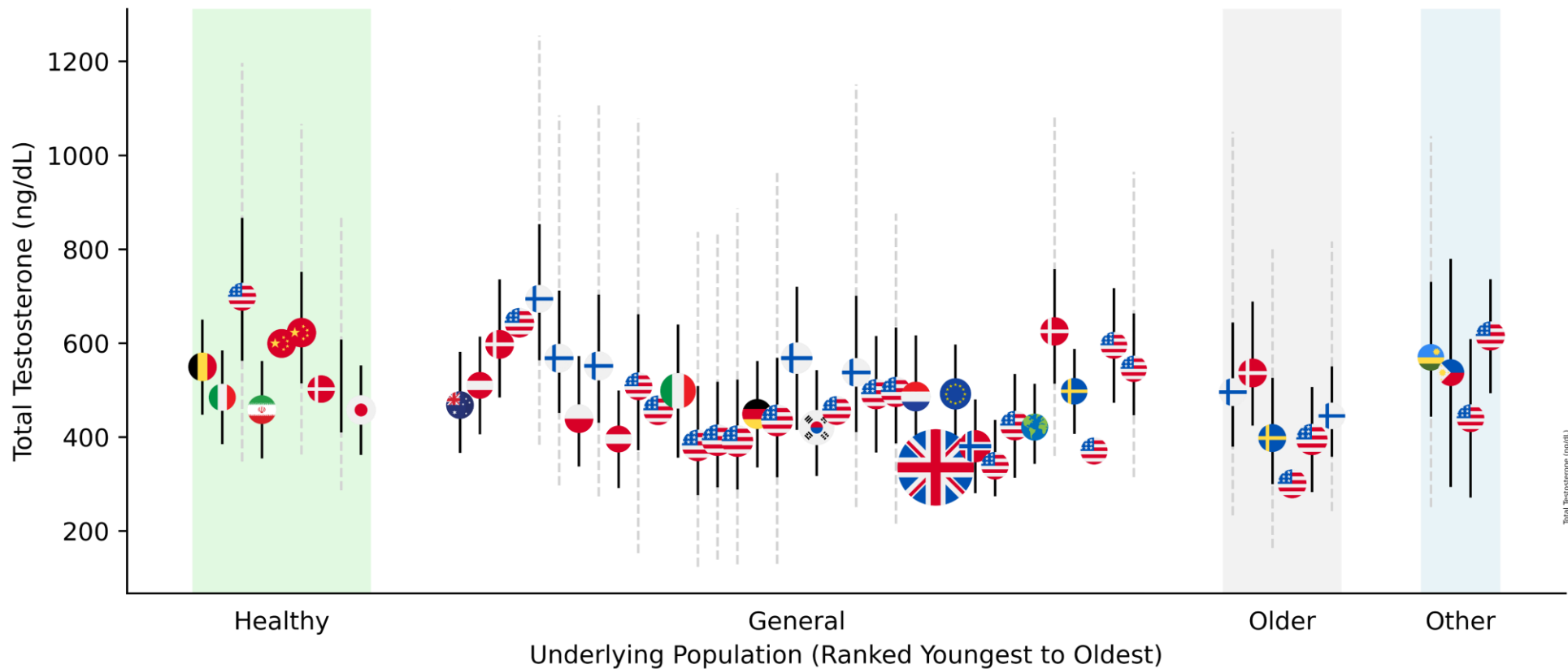
Mean Total T by Age

With T-Ranges and n

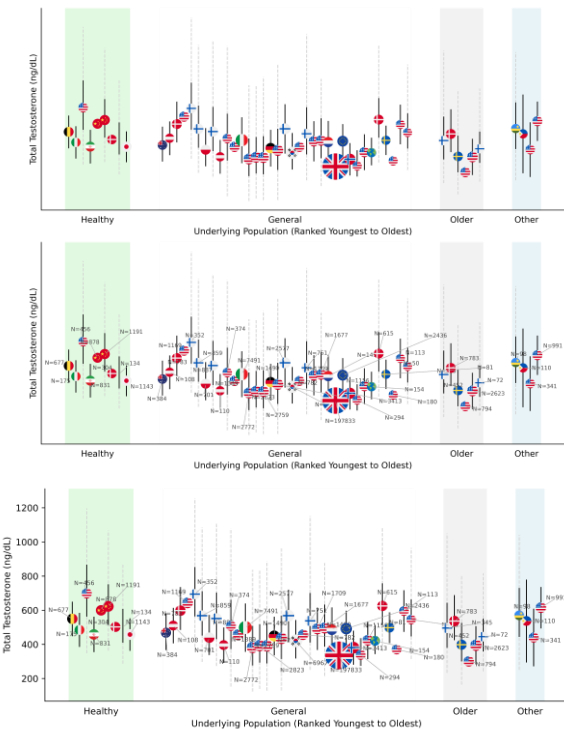


Mean Total T by Population

With T-Ranges



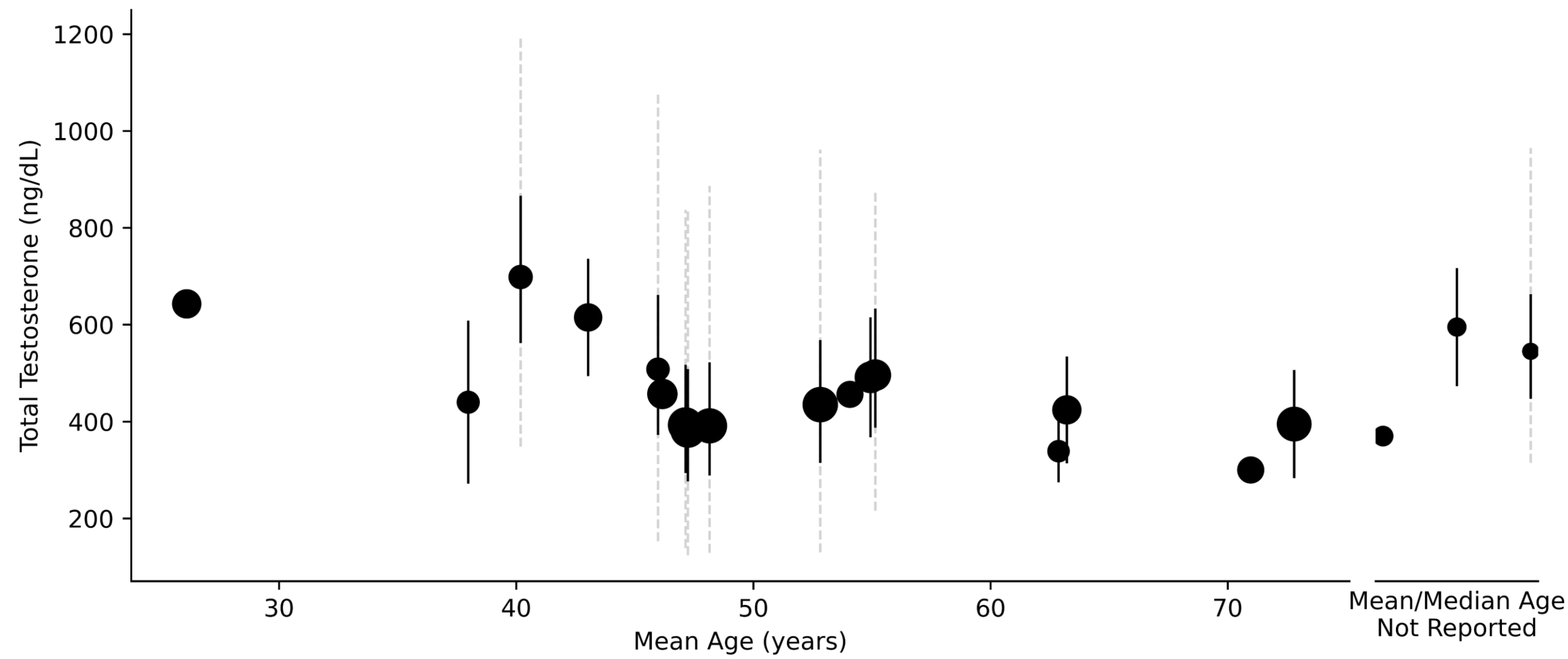
Other graphs with/without N and T-ranges



Testosterone range grouped by population included in study. The 'Other' groups from left to right consist of Rwandan medical students, Filipino university students and employees, American firefighters, and American Air Force veterans.

Mean Total T by Age

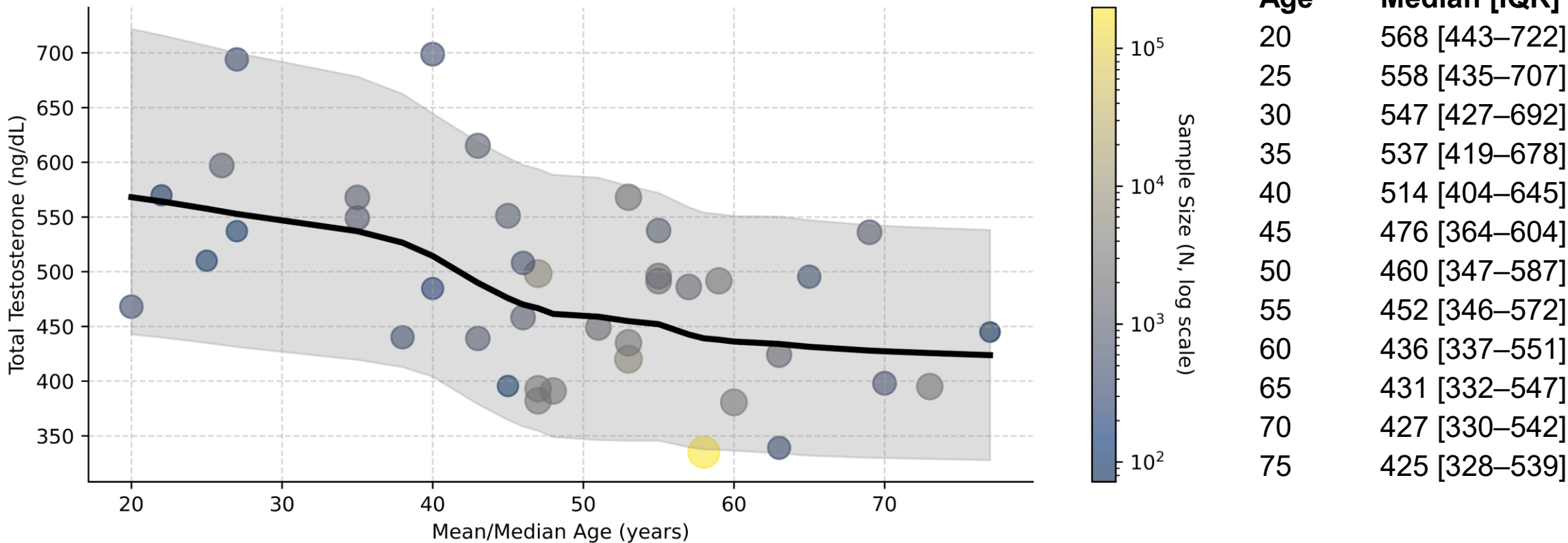
With T-Ranges, US Only (20 cohorts)



LOWESS Curve: Median total T by Age

Only cohorts reporting age (n=39) included

LOWESS Estimated Testosterone Median [IQR]



Aggregate median [IQR] serum testosterone: 490 [382 – 617] ng/dL